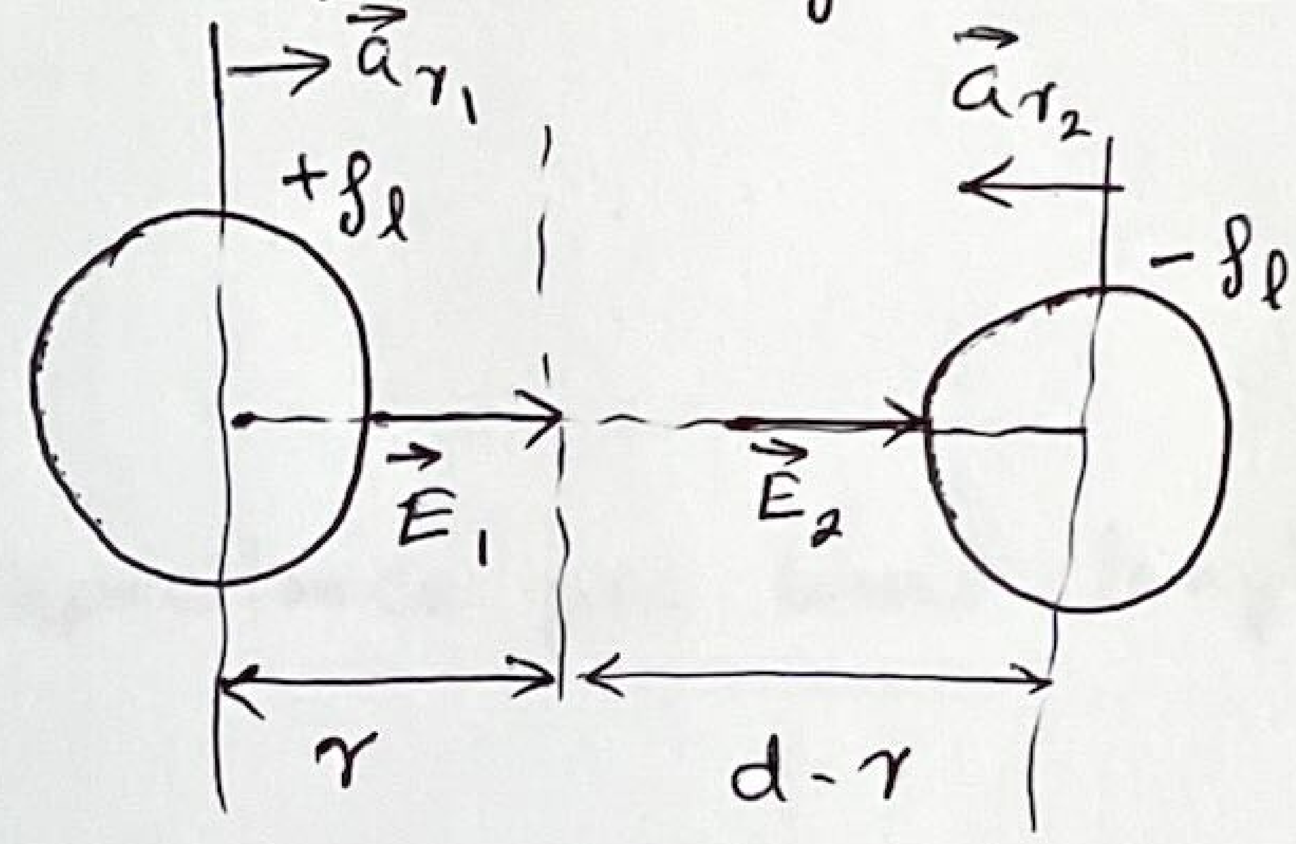
Question 1

(a) 5 marks

~~Let's assume~~ The line charge density ρ_ℓ of the conductor causes an electric field at an arbitary radius r, given as $\vec{E} = \frac{\rho_\ell}{2\pi\varepsilon_0 r}\vec{a}_r$

When $d \gg a$, the electric field of each conductor is not influenced by the elec. fld of the other.



The two electric flds at the two conductors are given as,

$$\vec{E}_1 = \frac{\rho_\ell}{2\pi\varepsilon_0 r_1}\vec{a}_{r_1}, \quad \vec{E}_2 = \frac{-\rho_\ell}{2\pi\varepsilon_0 r_2}\vec{a}_{r_2} \; ; \quad \text{Note } \vec{a}_{r_1} = -\vec{a}_{r_2}$$

(unit vectors)

Along the shortest line joining the centre, $\vec{E} = \vec{E}_1 + \vec{E}_2$

$$\vec{E} = \frac{\rho_\ell}{2\pi\varepsilon_0 r}\vec{a}_{r_1} + \frac{\rho_\ell}{2\pi\varepsilon_0 (d-r)}\vec{a}_{r_1}$$

~~marks~~ 3 marks

$$= \frac{\rho_\ell}{2\pi\varepsilon_0}\left[\frac{1}{r} + \frac{1}{(d-r)}\right]\vec{a}_{r_1}$$

$$|E(r)| = \frac{\rho_\ell}{2\pi\varepsilon_0}\left[\frac{1}{r} + \frac{1}{d-r}\right] \quad \text{—(1)}$$

Potential between the two conductors $V = \frac{\rho_\ell}{\pi\varepsilon_0}\ln\left(\frac{d}{a}\right)$

$$\Rightarrow \rho_\ell = \frac{\pi\varepsilon_0 V}{\ln(d/a)}$$

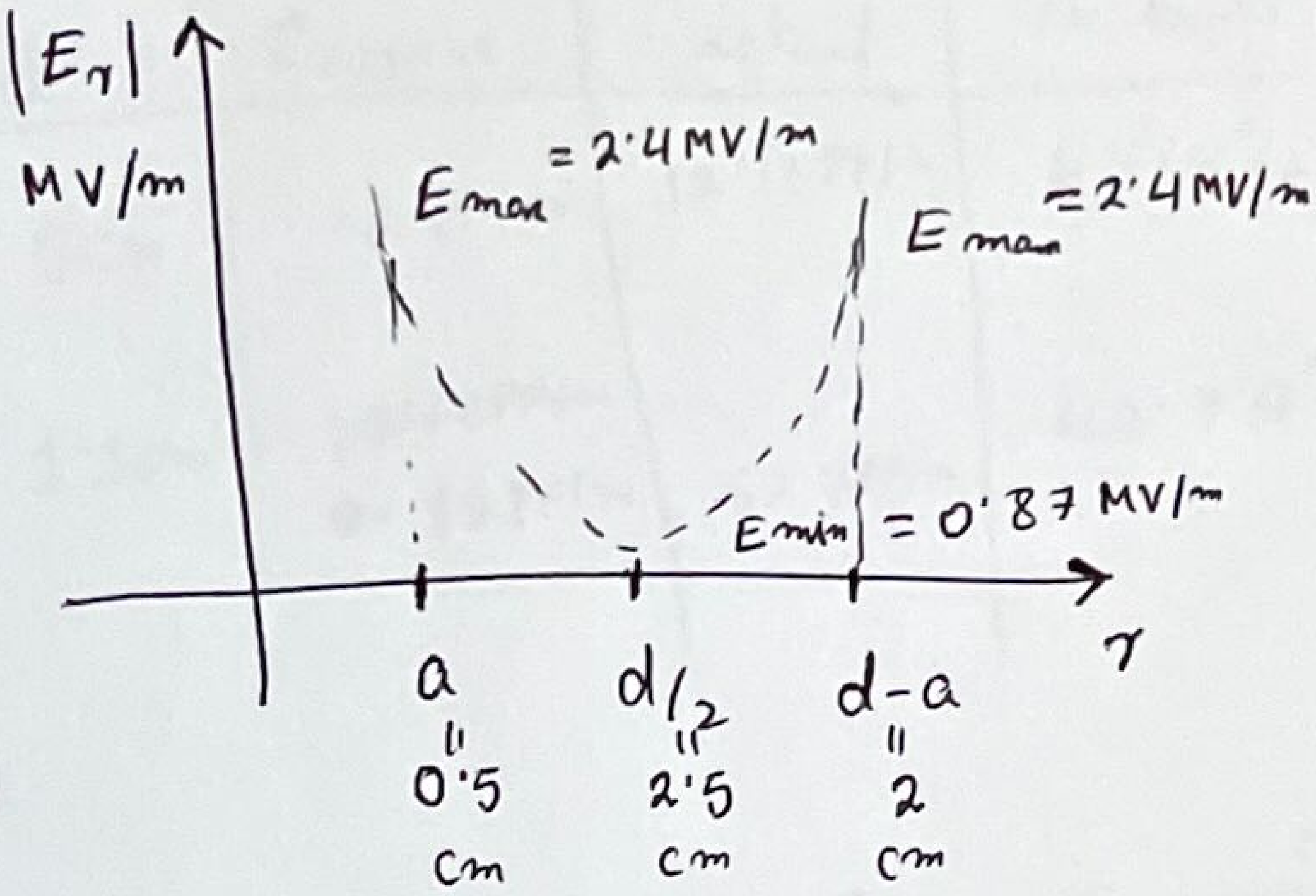
Sub. ρ_ℓ in (1),

2 marks

$$|E(r)| = \frac{1}{2\pi\varepsilon_0}\cdot\frac{\pi\varepsilon_0 V}{\ln(d/a)}\left[\frac{1}{r} + \frac{1}{d-r}\right]$$

$$= \frac{V}{2\ln(d/a)}\left[\frac{1}{r} + \frac{1}{d-r}\right]$$

(b) 3 Marks



1.5

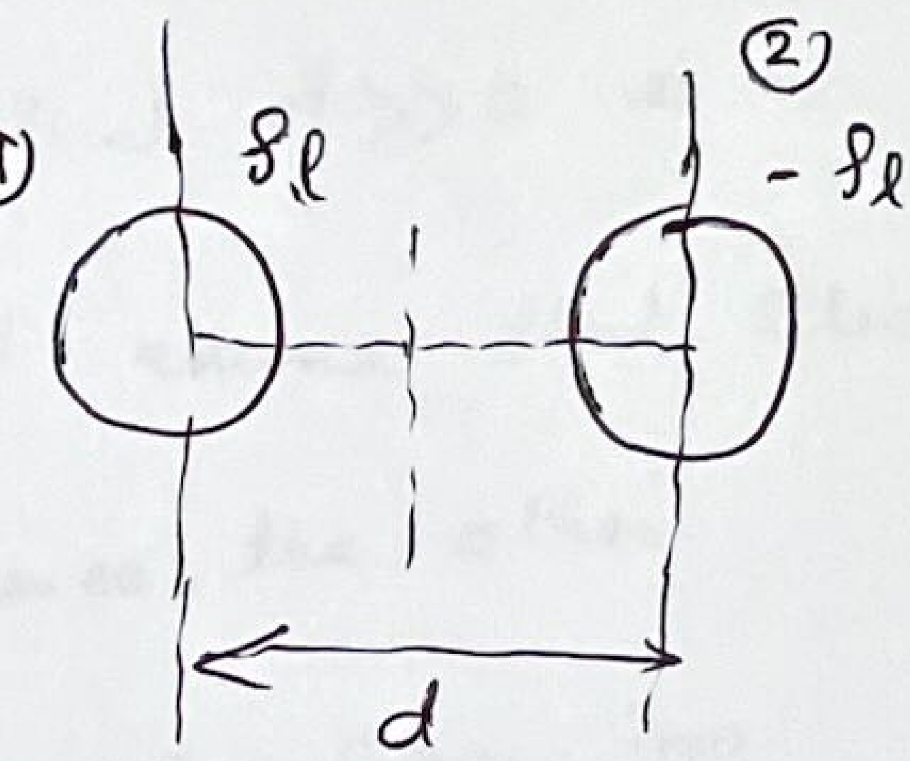
$$|E(r)| = \frac{V}{2\ln(d/a)}\left[\frac{1}{r} + \frac{1}{d-r}\right]$$

~~when~~

r	$\|E(r)\|$
a	2.4 MV/m
$d/2$	0.87 MV/m
$d-a$	2.4 MV/m

1.5

(c) Capacitance per unit length $C = \dfrac{\rho}{V_{12}}$



Potential at conductor 1, considering conductor 2 as the reference,

$$V_1 = -\int_{(d-a)}^{a} \frac{\rho_l}{2\pi\varepsilon_0 r}\,dr$$

2

$$= \frac{\rho_l}{2\pi\varepsilon_0}\ln\left(\frac{d-a}{a}\right)$$

Similarly, potential at conductor 2 due to $(-\rho_l)$ considering reference at conductor 1,

$$V_2 = -\int_{d-a}^{\cancel{d-a}} \frac{-\rho_l}{2\pi\varepsilon_0 r}\,dr$$

1

$$= -\frac{\rho_l}{2\pi\varepsilon_0}\ln\left(\frac{d-a}{a}\right)$$

$\therefore$ potential diff. betn the two conductors $V_{12} = V_1 - V_2 = \dfrac{\rho_l}{\pi\varepsilon_0}\ln\left(\dfrac{d-a}{a}\right)$

2

$$C = \frac{\rho_l}{\frac{\rho_l}{\pi\varepsilon_0}\ln\left(\frac{d-a}{a}\right)} = \frac{\pi\varepsilon_0}{\ln\left(\frac{d-a}{a}\right)} \approx \frac{\pi\varepsilon_0}{\ln(d/a)} \quad \left(\text{since } d \gg a\right)$$

(d)

d	C_{approx}	C_{actual}	% error	
5cm	12 PF/m	12.13 PF/m	0.44%	$\boxed{2}$
1.1cm	~~[illegible] PF/m~~ 35 PF/m	62.7 PF/m	43.74%	$\boxed{2}$

$$C_{approx} = \frac{\pi \varepsilon_0}{\ln(d/a)} , \quad C_{actual} = \frac{\pi \varepsilon_0}{\ln\left[\frac{d}{2a} + \sqrt{\left(\frac{d}{2a}\right)^2 - 1}\right]}$$

$$\% \text{ Error} = \frac{C_{actual} - C_{approx}}{C_{actual}} \times 100\%$$

(e) As d reduced to 1.1 cm, the assumption that $d >> a$ is no longer valid. As a result, we can ~~no~~ not assume that Elecl. fld of one conductor does not influence the other.

$\boxed{2}$ Thus, Gauss's law becomes inapplicable as we can no longer find a Gaussian surface on which $\vec{E}$ ~~is~~ of the conductor ~~normal~~ is normal.

Question 2 $\boxed{7 \text{ marks}}$

(a) The potential V varies only radially. hence the Laplace's equation is one dimensional and ordinary 2nd order differential equation.

$$\cancel{\frac{d^2V}{dr^2}} = \frac{1}{r}\frac{d}{dr}\left(r\frac{dV}{dr}\right) = 0 \rightarrow \frac{d}{dr}\left(r\frac{dV}{dr}\right) = 0 \quad 1$$

It can be solved by direct integration,

Intigrating once,

$$1 \qquad r\frac{dV}{dr} = k_1 \,, \quad k_1 : \text{integration constant.}$$

$$\Rightarrow \frac{dV}{dr} = \frac{k_1}{r}$$

Integrating again,

$$\cancel{V(r) = \frac{k_1}{r} + k_2}$$

$$1 \qquad V(r) = k_1 \ln(r) + k_2 \,, \quad k_2 : \text{Integration constant.}$$

k_1 & k_2 can be found from two boundary conditions,

$r = a, \quad V = V_0$

$r = b \quad V = 0$

$$k_1 \ln(a) + k_2 = V_0 \quad -(1)$$

$$k_1 \ln(b) + k_2 = 0 \quad -(2) \Rightarrow k_2 = -k_1 \ln(b)$$

Sub. in (1),

$$k_1 \ln(a/b) = V_0$$

$$k_1 = \frac{V_0}{\ln(a/b)} \quad 2$$

$$\therefore k_2 = -\frac{V_0}{\ln(a/b)} \ln(b)$$

$$\therefore V(r) = \frac{V_0}{\ln(a/b)}\left[\ln(r) - \ln(b)\right] \quad 1$$

$$\vec{E} = -\vec{\nabla} V$$

$$= \frac{-d}{dr}\left[\frac{V_0}{\ln(a/b)}\left(\ln(r) - \ln(a)\right)\right]$$

$$= \frac{-V_0}{\ln(a/b)} \frac{1}{r} \vec{a}_r \quad 2$$

(B) $V_0 = 60\,V$,

E_{max} @ $r = a$ $\quad E_{max} = -\frac{V_0}{\ln(a/b)} \frac{1}{a} = 52\ kV/m \quad 2$

E_{min} @ $r = b$ $\quad E_{min} = \frac{V_0}{\ln(a/b)} \frac{1}{b} = 5.2\ kV/m \quad 2$

(C) $\vec{E} = -\frac{V_0}{\ln(a/b)} \frac{1}{r} \vec{a}_r$

Using points form of ohm's Law, $\vec{J} = \sigma \vec{E}$ $\quad 2$

$$= -\frac{\sigma V_0}{\ln(a/b)} \frac{1}{r} \vec{a}_r$$

Current $I_{lk} = J.S$

$$= \frac{\sigma V_0}{\ln(a/b)} \frac{1}{\cancel{r}} (2\pi \cancel{r}.h) \quad 2$$

$$= \frac{-\sigma V_0 (2\pi h)}{\ln(a/b)}$$

Resistance $R_{lk} = \frac{V_0}{I} = \frac{-1}{2\pi h \sigma} \ln(a/b)$

$$= \frac{-1}{2\pi h \sigma} (\ln a - \ln b)$$

2

$$= \frac{1}{2\pi h \sigma} \ln(b/a)$$

(d) $I_{lk} = -\dfrac{\sigma V_0 (2\pi h)}{\ln(a/b)} = 0.82\,mA$ 2

Power loss $= V_0 I_{lk} = 0.049\,W \approx 49\,mW$ 2

or Power loss $= I_{Lk}^2 R_{Lk}$, $R_{lk} = \dfrac{1}{2\pi h \sigma} \ln(b/a)$

$= 0.73\,m\Omega$

or Power loss $= \dfrac{V_0^2}{R_{lk}}$.

Question 3

(a) 'Virtual displacement' assumes the stored energy is spent to produce a displacement.

Mech. work done dW_{mech} = ~~Force~~ Force, displacement. $= \vec{F}.d\vec{l}$

$dW_{mech} = \vec{F}.d\vec{l} = dW_s - dW_m$

where dW_s : Source energy

dW_m : Energy stored in the magnetic field.

~~6~~

4

If no source energy, $\vec{F}.d\vec{l} = -dW_m$

$$\vec{F} = -\frac{dW_m}{d\vec{l}}$$

For a 3-dimensional ~~quantity~~, $\vec{F} = -\vec{\nabla} W_m$

energy ~~force~~ and displacement function,

$$W_m = \frac{1}{2} L I^2 = \frac{1}{2} \lambda I, \quad \because L = \frac{\lambda}{I}$$

Force can be found from gradient of stored energy W_m assuming either current or flux linkage constant.

If constant current is assumed, ~~then~~ it means a source ~~are~~ is present and magnetic energy is refilled ~~to~~ by

the source to do the work of virtual displacement.
i.e force is positive gradient of the stored energy.

$$\therefore \quad \vec{F} = +\vec{\nabla} W_m \Big|_{\text{constant current.}}$$

(b) To find force, we need the stored energy W_m.

$$W_m = \frac{1}{2} \lambda I$$

$$\boxed{2}$$

$$\lambda = \int \vec{B} \cdot d\vec{s} \quad \text{and} \quad I = \oint \vec{H} \cdot d\vec{l}$$

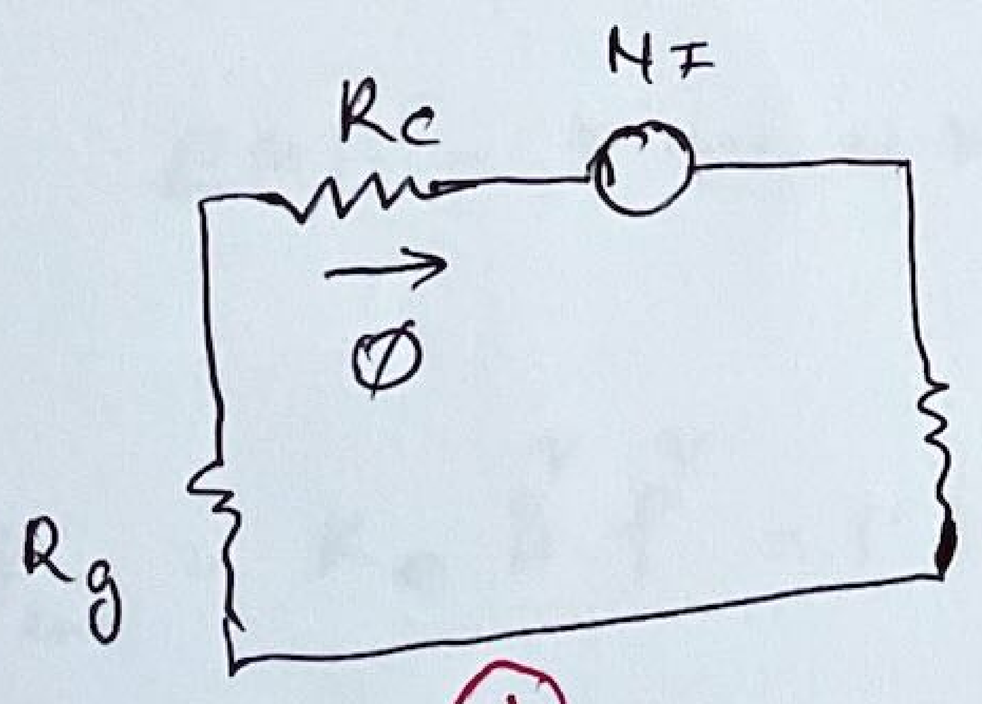
Using these relation, $W_m = \left(\frac{1}{2} \vec{B} \cdot \vec{H}\right) \times \text{Volume} = \frac{B^2}{2\mu_0} \times \text{Vol.}$

Energy stored in the airgap $W_m = \frac{B^2}{2\mu_0} \times \text{Volume} = \frac{B^2}{2\mu_0} (S \cdot l_g)$

Due to the force air gap length l_g varies. $\therefore W_m = \frac{B^2}{2\mu_0} (S \cdot y)$

Assuming uniform flux density, no leakage and fringing,

a magnetic circuit of the device can be drawn as



1

$$R_c = \frac{l_c}{\mu_c S_c} = 7.76 \times 10^5 \quad 1$$

$$R_g = \frac{l_g}{\mu_0 S_g} = 9.95 \times 10^6 \quad 1$$

no fringing $\Rightarrow S_c = S_g$

$$l_c = (400 - 10) = 390\,mm$$

$$l_g = 5\,mm$$

$$S_c = S_g = 400\,mm^2$$

$N = 5000$
$I = 0.1A$

$$\phi = \frac{NI}{2R_g + R_c} = 2.42 \times 10^{-5}\ Wb \quad 1$$

$$B = \frac{\phi}{S} = 0.0605\ T \quad 1$$

Force is in the y direction only.

$$\vec{F}_y = \left[\frac{d}{dy}\left(\frac{B^2}{2\mu_0} S . y\right)\right](-\vec{a}_y) \quad \text{(negative y direction)}$$

$$= \frac{B^2}{2\mu_0} . S = \frac{0.0605^2}{2\mu_0} . S = 0.582\ N \quad 2$$

*' [If total force of the system is required then energy stored in two gap should be considered. i.e. $W_m = \left(\frac{B^2}{\mu_0} S . y\right)$

$$\vec{F}_y = -\frac{B^2}{\mu_0} S\, \vec{a}_y = 1.14\ N$$]

This is also correct.

(c) Inductance $L = \frac{N^2}{R_{Total}}$, $R_{Tot} = 2R_g + R_c$

$= 1.21\ H$ 2

(d) For a.c supply ϕ found in (b) is the peak value.

$$EMF = 4.44 N f \phi = 26.87\ V\ (rms) \quad 2$$

(e) $P_{Loss} = k_e B^2 f^2 = 1.1$ Watts. 2

Using ~~too~~ thin lamination to make the core can reduce eddy current loss. 1